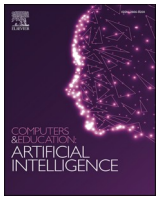
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Exploring students' experience of ChatGPT in STEM education

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ABSTRACT

The rapid advances in AI technologies showed a disruptive potential in educational practices, presenting new challenges and generating new opportunities. This phenomenon has been exacerbated since the release of ChatGPT in 2022, which has permanently transformed various educational activities and sparked widespread scientific interest. Research suggests that ChatGPT can help students navigate the complexities of STEM subjects. However, only a few studies have directed attention to the use of ChatGPT in STEM subjects in upper secondary education. With the purpose of addressing this gap, the aim of this study is to explore how students experience ChatGPT for their STEM studies, encompassing their usage, perceptions, and general knowledge about this technology. Using a mixed methods approach, the data collected included a survey and semi-structured interviews involving upper secondary students. The results show widespread adoption of ChatGPT across STEM subjects among participants, particularly in biology and especially as a tool to support the understanding of concepts. Although students exhibited limited knowledge of AI, they demonstrated some effective prompting strategies to generate relevant content and tackle potential inaccuracies and hallucinations. The findings in this paper provide insights to support the exploration of students' experiences of ChatGPT, presenting relevant topics to further research the applications of these AI technologies within STEM subjects, given their importance for future societal development.

1. Introduction

Artificial intelligence in education (AIED) refers to the application and research of AI technologies in educational contexts, exploring their potential to support learning and reshape educational practices (Hwang et al., 2020; Xu & Ouyang, 2022; Zafari et al., 2022). Within this broader field, AI STEM (Science, Technology, Engineering, and Mathematics) has emerged as an area of research over the past decade. This specific focus on STEM education highlights the potential of AI technologies to enhance teaching and learning in these critical subjects, which are increasingly recognised as vital for addressing the growing demand for a skilled STEM workforce.

The importance of STEM education has increased in recent years, prompting various educational systems to prioritise STEM subjects and programs. This shift is driven by the anticipated shortage of STEM professionals (Caprile et al., 2015), a concern that has led to strategic initiatives in many countries. For instance, in Sweden, the Department of Education issued a press release in 2024 outlining a strategy to encourage more students to enrol in STEM programs, addressing this emerging workforce gap (Utbildningsdepartementet, 2024). Additionally, recent studies provide evidence of AI's potential to support critical skills such as computational thinking and problem-solving, which are essential for STEM education (Xu & Ouyang, 2022; Yang et al., 2024). These findings emphasise the need for further empirical research on the role of AI in STEM education.

The release of ChatGPT in 2022 further underscores the transformative impact of AI in education. Immediately after its first release to the public, even the CEO of OpenAI (the company that released ChatGPT) warned of the potential risks of overly relying on ChatGPT, stressing that its generated texts give only the impression of being great while building on unstable grounds: "ChatGPT is incredibly limited but good enough at some things to create a misleading impression of greatness. It's a mistake to be relying on it for anything important right now" (Altman, 2022).

Much of the existing literature on Chatbots based on Large Language Models (LLM Chatbots), such as ChatGPT in education, focuses primarily on higher education in general (Albadarin et al., 2024; Ansari et al., 2024) and not on STEM education. This focus is mainly due to the technology's proficiency in generating written content, which has significant implications for home assignments and examinations in general

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(Bower et al., 2024; Dwivedi et al., 2023; Rahman & Watanobe, 2023; Stojanov et al., 2024). Although limited, current research highlights various opportunities offered by LLM chatbots such as ChatGPT, for STEM learning. For instance, a study by Ng et al. (2024) suggests that this technology can be used to support students' self-regulated learning in science education by assisting them in generating new goals and developing strategies to enhance their studies. To investigate these opportunities further, this paper explores how students perceive and use LLM chatbots like ChatGPT specifically for STEM subjects within the context of upper secondary school education. Additionally, this paper explores students' perceived knowledge about AI and prompting strategies. A single broad research question was formulated to capture this study's explorative nature.

- How do students experience ChatGPT for their STEM studies?

Although this study did not focus solely on ChatGPT but included all currently available LLM chatbots, such as Gemini and Copilot, both the research question and results refer to the technology as ChatGPT. This approach was chosen due to the predominance of ChatGPT's adoption reflected in the data.

2. Background

2.1. ChatGPT in education

The disruptive potential of generative AI technologies such as ChatGPT in the school context has significantly altered how various educational activities are conducted, particularly in the realm of assessment. The widespread use of these tools emphasises the importance of developing new types of "generative AI-proof" exams designed to assess students' work in a way that accounts for their use of AI technologies and ensures the integrity of examination processes (Larsen, 2023; Niloy et al., 2024). Furthermore, the potential of the technology to enable cheating resulted in the need for educational institutions to establish explicit norms for using tools such as ChatGPT and design new assessments that account for these emerging technologies (Adeshola & Adepoju, 2023; Dwivedi et al., 2023). This approach aims to deter AI cheating and ensure academic integrity without solely relying on students' moral integrity.

Despite the numerous concerns, ChatGPT has shown the potential to be a practical tool for teachers to systematically assess and provide feedback to students (Kasneci et al., 2023; Meyer et al., 2024), although the long-term effects of this specific application are yet to be observed. Preliminary results show that ChatGPT can also be an effective tool for generating classroom materials by reframing problems and generating content for learning activities (Einarsson et al., 2024). Its ability to quickly generate educational materials could be an important resource for teachers to be able to personalise the content of activities to specific students, achieving a higher level of content personalisation to address students' needs. Due to heavy workload, these activities usually require a substantial amount of time that is not always available for teachers (Baidoo-Anu & Owusu Ansah, 2023).

The implications of adopting AI tools such as ChatGPT have also been explored from an epistemic perspective, with concerns raised about overreliance on these technologies as sources of information (Chavanayarn, 2023; Darvishi et al., 2024; Dwivedi et al., 2023; Koskinen, 2024). Key concerns include the inability of these tools to truly "understand," the black-box nature of their operations, and the potential harm to the learning process by impairing the development of critical thinking skills (Dwivedi et al., 2023; Wu, 2024). Overreliance has also been discussed in decision-making contexts, highlighting the risks of bias and the potential for reduced originality in academic work (Halaweh, 2023; Lin et al., 2023).

Several studies explored the implications of AI for STEM education, notably the work of Xu and Ouyang (2022) and Yang et al. (2024), although these studies do not specifically focus on ChatGPT. Among the findings presented, these technologies—beyond supporting core STEM skills—are argued to have the potential to transform learning practices by shifting toward more student-centred learning environments. Such an approach could potentially support students in navigating the complexity of STEM subjects. Additionally, a few papers have explored ChatGPT within specific STEM subjects, such as Liang et al. (2023) where ChatGPT's potential implications for physics education is explored. In this paper, the authors suggest that ChatGPT can support students in their studies while also creating an environment where they can explore concepts without the fear of judgment. In this context, students can ask questions they might otherwise consider out of place, helping to clarify underlying misconceptions.

2.2. Prompt engineering

One of the characteristics of LLM chatbots, such as ChatGPT, is their tendency to always provide an answer, even when the AI lacks the capacity to reliably produce a correct response or when the prompt does not contain a clear question (Theophilou et al., 2023). This, among other things, can lead to issues with the accuracy of the generated text. However, the accuracy can be improved through specific prompting strategies, which have been shown to significantly enhance ChatGPT's performance, particularly in tasks requiring logical analysis or problem-solving skills (Wang et al., 2023, pp. 1–13). The emerging concept of prompt engineering refers to these strategies designed to increase the accuracy and usefulness of ChatGPT, especially in STEM-related subjects like physics, where logic and problem-solving are critical.

Cain (2024) raises three main components that characterise effective, prompt engineering in educational contexts: (a) content knowledge, (b) critical thinking, and (c) iterative design approach. The role of content knowledge in prompt engineering is primarily to assess and evaluate text generated by AI to enable an iterative approach for refining and eventually fine-tuning the initial prompt. Critical thinking has a similar role as a mindset to support the evaluation process and further contribute to the iterative approach as the last main component (Cain, 2024).

An effective strategy for enhancing the accuracy of ChatGPT in solving logic problems, as demonstrated by Wei et al. (2022), involves presenting the Chatbot with a similar problem and its step-by-step solution before introducing the actual logic problem. This approach has been shown to significantly improve ChatGPT's ability to solve problems requiring logical reasoning. Teaching students well-conceived prompting strategies like this may influence their perceptions of interactions with ChatGPT, especially regarding the perceived usefulness of the generated response (Tassoti, 2024). Moreover, mastering a broad range of prompting strategies could make these interactions more natural, enjoyable, and effective (Theophilou et al., 2023). Other prompting strategies involve contextualising prompts or specifying the domain related to the inquiry to enhance relevance to the subject to which the activity is related (Polverini & Gregorcic, 2024).

2.3. AI literacy

LLM chatbots such as ChatGPT, while effective for generating text that appears man-made, come with various issues that must be addressed in educational settings to encourage careful and informed use of AI. Recently, the concept of AI literacy has gained attention in scholarly research. It is a framework that attempts to conceptualise the skills necessary to use the vast array of AI systems publicly available effectively and responsibly (Ng et al., 2021a). One of the first and widely adopted definitions stems from a conference paper published by Long and Magerko (2020) where the authors define AI literacy as "a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a

tool online, at home, and in the workplace" (p. 2). Long and Magerko identifies 17 key competencies that characterise AI literacy, where the concept is not specifically observed from an educational perspective but instead as a general phenomenon. During the past years, different papers further conceptualised AI literacy resulting in some form of agreement regarding its key constructs. In these conceptualising papers, the main competencies identified are (a) recognise, (b) know and understand, (c) use and apply, (d) evaluate, (e) create, and (f) navigate ethically (Almatrafi et al., 2024).

A recently accepted paper by Annapureddy et al. (2024) further contextualises the concept of AI literacy specifically in relation to Generative AI to address the widespread phenomenon of Large Language Models, proposing the idea of Generative AI literacy. Generative AI technologies bring to the table the need to discuss competencies that are not explicitly addressed in the general AI literacy models, such as prompt engineering strategies and the ability to detect AI-generated content. Annapureddy et al. (2024) present a set of competencies that build upon previous conceptualisations of AI literacy but provide greater specificity for the context of generative AI. The authors further stress the importance of understanding and exploring the concept of AI literacy in relation to generative AI technologies due to their impact on our everyday lives.

3. Methodology

This study is characterised by a mixed methods approach, chosen to facilitate a descriptive analysis of empirical data, thereby providing an overview of students' attitudes towards and descriptions of their use of ChatGPT. The methodological approach was chosen not to draw inferential conclusions but rather to facilitate the collection of rich empirical data. This strategy aims to comprehensively describe the phenomenon of interest, focusing on a detailed understanding rather than predictive or causal inferences. The study is inspired by the Explanatory Sequential Design presented in Fig. 1, articulated in two phases as described by Creswell and Creswell (2023).

In the initial phase, a survey consisting of close-ended questions was conducted. The quantitative data collected from this survey was first analysed to offer preliminary insights into students' experience of ChatGPT STEM education. This analysis aimed to explore how students perceived ChatGPT in STEM education and to gather insights into their AI knowledge. The survey was anonymised to minimise the collection of personal data and to increase the response rate.

Following the completion of phase one, with the insights obtained, the study progressed to the second phase. The data from the first phase supported the formulation of interview questions, adopting an informed approach based on the initial data analysis. Subsequently, the interviews were transcribed, coded, and analysed following the thematic analysis framework proposed by Braun and Clarke (2006). Finally, this study integrated the results from the thematic analysis with the descriptive statistical data collected initially. This comprehensive approach, as visualised in Fig. 2 through a schematic representation of the research process, was chosen to construct a nuanced presentation of students' interactions, perceptions and attitudes towards ChatGPT, specifically in the context of STEM education.

3.1. Adaptation of the AI literacy model

This section presents the AI literacy model adopted to structure the quantitative phase of the study. This model was constructed based on the work of Kong et al. (2024) and Annapureddy et al. (2024). Due to the nature of this study, which does not focus on AI in general but rather specifically on ChatGPT contextualised in STEM education, the more general definition of AI literacy, such as the one presented by Long and Magerko (2020) see section 2.3., becomes too broad to integrate the concept of AI literacy within this paper. For this reason, the framework proposed by Kong et al. (2024) together with the competencies presented by Annapureddy et al. (2024), offers a more suited conceptual tool to address the gathered data. Kong et al. (2024) structure the concept of AI literacy in four dimensions inspired by previous conceptualisation attempts that focused on AI literacy in the past few years as follows: (1) *Cognitive dimensions* (the understanding of AI concepts); (2) *Metacognitive dimension* (Knowledge required to apply and use AI tools); (3) *Affective dimension* (perceived psychological readiness to use AI for different purposes); and (4) *Social dimension* (Ethical aspects related to the use of AI).

Although this model concentrates explicitly on problem-solving, the presented dimensions can be adapted to investigate students' AI literacy in relation to generative AI. Dimension 1, 2 and 4 are heavily influenced by the earlier AI literacy conceptualisation studies such as the works of Long and Magerko (2020) and Ng et al. (2021a, 2021b), while Dimension 3 is, on the other hand, the result of the analysis of more recent papers and focuses on user perceptions. This dimension relates to the perceived self-efficacy, or the user's readiness to adopt AI technologies for a specific purpose or activity. The four categories are further recontextualised for this study in the following way:

1. *Cognitive Dimension:* Understanding fundamental concepts behind LLM chatbot technologies.
2. *Metacognitive Dimension:* Capability to use ChatGPT for STEM-related studies
3. *Affective Dimension:* Students' perceived readiness to use ChatGPT for STEM-related studies.
4. *Social Dimension:* The ethical and privacy implications of using ChatGPT and the rules that regulate its usage for schoolwork.

The competencies presented by Annapureddy et al. (2024) serve on the other hand as a tool to capture and structure students' AI literacy specifically within the context of generative AI technologies, in the case of this study, ChatGPT. The only dimension that is least represented within these competencies is the Affective dimension. This is likely a consequence of the fact that its definition stems from different conceptualising attempts than the other three that share more common references with the work of Annapureddy et al. (2024). The combined model of Kong et al. (2024) and Annapureddy et al. (2024) is presented in Table 1 which shows how the competencies are organised as constructs of the AI literacy dimensions and represents the model adopted while formulating the survey questions and presenting the quantitative data. Furthermore, a new competence was added: *Perceived readiness to use generative AI tools,* in order to address the underrepresented third dimension within the competencies suggested Annapureddy et al. (2024).

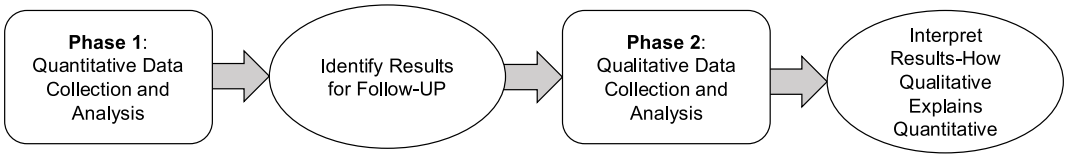


Fig. 1. Explanatory Sequential Design
Note. Adapted from Creswell & Creswell, 2023, p. 236.

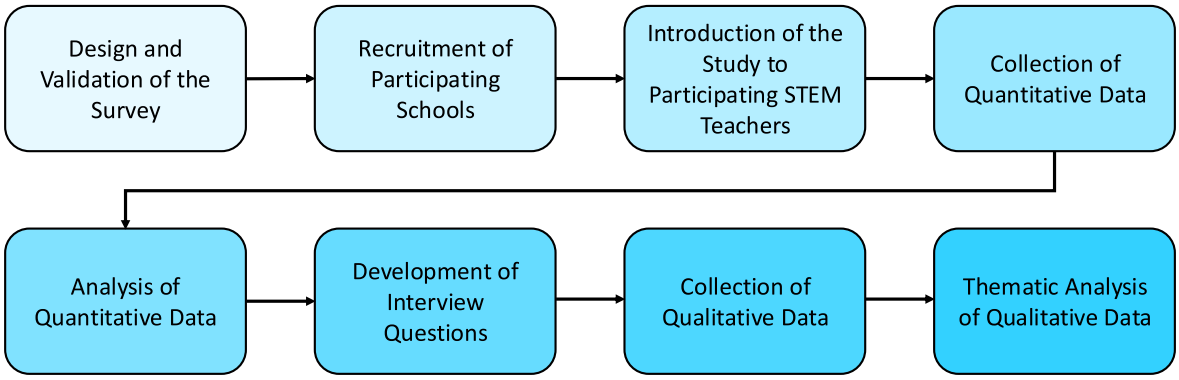


Fig. 2. Research process.

Table 1
Competencies as constructs of the AI literacy dimensions.

Dimensions	Competencies
Cognitive	Basic AI literacy, Knowledge of generative AI models, Knowledge of Capacities and limitations of Generative AI tools
Metacognitive	Skill to use Generative AI tools, Ability to detect AI generated content, Ability to assess the output of generative AI tools, Skill in prompting generative AI tools, Ability to program and finetune, Knowledge of the context where generative AI is used
Affective	Ability to continuously learn[a], Perceived readiness to use generative AI tools[b]
Social	Knowledge of ethical implications, Knowledge of legal aspects

Note. Contextualisation of Annapureddy et al. (2024) competencies within Kong et al. (2024) AI literacy dimensions.
[a]Observed as students' attitude towards learning about AI.
[b]Not part of the 12 competencies.

Specifically, in this study, the competencies structured in Table 1 are mainly observed within STEM education to better serve as a discussion tool to address the gathered empirical data.

3.2. Recruitment and participants

During the recruitment phase, the principals of 24 upper secondary schools were contacted via email. The schools were selected based on two criteria: (a) their geographical location and (b) their offering of either the Natural Science or the Technology programme. The principals were initially contacted to obtain formal approval for the schools' participation. Subsequently, the project was presented to the teachers within the relevant programme to recruit participants and clarify aspects of the study. All schools meeting these criteria were contacted, with geographical proximity being an important consideration due to the study being part of a larger project with strict logistical requirements. A collaboration was established with only two municipal schools and one private school, all participating in both phases of this study. Only one of the municipal schools offered both programmes, while the remaining ones only offered the Natural Science programme. At the time this study was conducted, the total number of students enrolled in the Natural Science and Technology programmes at the school offering both was 280 and 260, respectively. In contrast, the remaining schools, which offered only the Natural Science programme, had a total of 225 and 150 students enrolled.

The quantitative phase involved a survey administered to students ($n = 453$) in their first or second year of upper secondary school, enrolled in either the Natural Science or the Technology programme. These programs were chosen to ensure that participants were exposed to various STEM subjects, such as mathematics, physics, chemistry, and biology. The survey was presented to students by their classroom teachers and conducted during class time. Each school was instructed to administer the survey to 150 students, except for the school offering both programs, which presented the survey to 150 students from each programme. This brought the total number of students targeted for the survey to 600, with 453 responses collected, resulting in a response rate of 75.5%.

Participants for the qualitative interviews were selected from the classes that completed the survey in the first phase. Students were informed by their teachers about the opportunity to participate in a follow-up interview focused on the use of generative AI in STEM education. Among the volunteering students, six were selected from the school offering both programmes (three students per programme), and three students from each of the two remaining schools were chosen by their classroom teachers and subsequently interviewed. The data collected from the 12 interviews was deemed sufficient for the purposes of the thematic analysis and addressing the research question, so no further interviews were conducted. Although allowing teachers to select participants from the volunteering students may have introduced some bias into the sampling process, this approach was chosen to minimise disruption to lesson time. Conducting interviews during class could have otherwise negatively impacted the participants' education. Additionally, randomly selecting students from the volunteers might have increased the risk of over-representing those who were primarily motivated to skip the particular lesson during which the interviews were conducted.

3.3. Phase 1 - quantitative data collection

The survey created for this study included questions employing a 5-point frequency scale alongside 5-point Likert scale questions for assessing responses on unipolar and bipolar scales, respectively. Furthermore, it featured two category-based items designed to capture non-ordinal data. The choice of diverse scales stems from the exploratory nature of the study's initial phase.

The AI literacy model presented in Section 3.1 served as a guideline to inform the creation process and to identify several survey items. Additional items were selected to explore students' use of ChatGPT more generally, beyond the concept of AI literacy. To validate the survey items, each item was initially discussed among the authors to avoid the inclusion of vague or leading questions. Following this, a pilot survey was conducted with 10 participants, who provided written and oral feedback on the clarity of the survey items and highlighted potential issues related to their validity. Based on this feedback, the survey items were subsequently revised and finalised.

While the data collected is quantitative and responses to questions close-ended, the objective was not to employ inferential statistics for hypothesis testing. Instead, the goal was to organise the collected data to facilitate the subsequent phase of interviews. This adoption of a multi-scale approach aimed at enhancing the validity of each question and offering descriptive statistics based on rich and diversified data.

Furthermore, each item within a scale was described with words to facilitate a more universal understanding of the items across different participants (Krosnick, 2018).

The categorisation of the Likert scale as strictly ordinal or continuous remains a subject of debate (Harpe, 2015); however, this distinction is of minimal concern for this study, as no inferential statistics are to be derived from the Likert items, which will be treated as ordinal data. This approach precludes any assumptions of a continuous distribution among the items. Instead, the analysis will solely present the distribution of responses in percentages for each item, thereby facilitating comparative analysis rather than assuming any form of inference. An illustrative frequency question relevant to this study is translated and presented as follows:

During the past month, how often did you use AI tools like ChatGPT to help you with your studies for scientific subjects?

- I didn't use any AI tools
- 1–2 times
- 3–5 times
- 6–10 times
- >10 times

According to Krosnick (2018), the optimal reliability of survey responses for this type of question is achieved using a 5-point scale. This scale is characterised by having a midpoint; however, it does not signify neutrality. Instead, it represents a gradation of the frequency of the described activity, commencing from a theoretical zero point, indicating the complete absence of the activity. For this reason, within the context of this study, unipolar questions were denoted by a 5-point frequency scale.

In contrast, bipolar Likert scale questions assess two opposed attitudes, characteristics, or perceptions. These questions are usually characterised by a 5 or 7-point Likert scale, featuring a clearly defined middle option such as "neutral" or "neither agree nor disagree" (Krosnick, 2018). This configuration is designed to capture reliable data by allowing respondents the option of neutrality while preventing the ambiguity that may arise from more extensive scales with unclear item delineations. For this study, a 5-point Likert scale was adopted. Below is a contextualised example of a bipolar question translated into English:

AI tools like ChatGPT help me understand concepts in STEM subjects.

- Strongly disagree (ChatGPT hinders my understanding)
- Somewhat disagree (ChatGPT offers limited help)
- Neither agree nor disagree (ChatGPT is neutral for my understanding)
- Somewhat agree (ChatGPT provides some assistance for understanding)
- Strongly agree (ChatGPT greatly supports my understanding)

Finally, some of the questions related to students' use of the technology included an additional option to answer "I don't use any AI tool for my studies." This was done to increase the validity of the survey responses, considering the varying levels of ChatGPT use among participants.

3.4. Phase 2 - qualitative data collection

The qualitative phase of this study involved semi-structured interviews conducted with 12 different students. Each interview lasted between 15 and 20 min and was recorded to facilitate subsequent analysis. Semi-structured interviews were chosen to ensure flexibility due to the exploratory nature of the study. The interviews were conducted by a member of our research group who had prior experience with semi-structured interviews, and the interview guide was developed after the initial analysis of the survey data, following the structured approach outlined Bryman (2016). The process began by selecting interview topics that were closely aligned with the results of the quantitative phase and directly related to the research question. This step was essential to ensure that the topics provided sufficient coverage for a comprehensive data collection. Following the selection of topics, preliminary interview questions were drafted. The coauthors then collectively reviewed these initial questions through a dialogical approach, which involved open discussions and collaborative refinement. This iterative process allowed for multiple rounds of revision, ensuring the questions were clear and aligned with the study's objectives. After several iterations, the interview questions were finalised, and the interview guide was completed. The dialogical and iterative approach was chosen to minimise the risk of leading questions and to avoid the influence of individual preconceptions, ensuring that the questions were objective and relevant (Bryman, 2016).

To illustrate how the quantitative data informed the second phase of the study, a specific example is provided to demonstrate how the results of a particular survey question guided the creation of the interview guide. The survey revealed that over 60% of the participants agreed with the statement: "AI tools such as ChatGPT help me understand concepts in STEM subjects like mathematics, technology, physics, chemistry and biology." This finding was directly incorporated into the interview guide with direct, probing and eventually follow-up questions to investigate this phenomenon in greater depth. The interviews aimed to gather qualitative data to understand the reasons behind the perceived usefulness of ChatGPT in this context.

3.5. Analysis

The quantitative data was analysed using the digital tool SPSS to generate descriptive statistics. While not the primary focus of the study, some correlations were also explored. For instance, the relation between students' perceived ability to discern ChatGPT-generated hallucinations and their specific knowledge about ChatGPT was examined. In addition to presenting descriptive statistics, the analysis primarily aimed to inform the qualitative phase of the study. The bulk of the data was analysed by observing percentages, with particular attention given to survey questions that exhibited substantially skewed responses such as the example previously mentioned in 3.4. Furthermore, multiple diagrams and tables were generated to facilitate the sharing and discussion of data among the coauthors of this paper. These visual tools were essential in highlighting interesting aspects of the data and suggesting potential correlations to explore further. This part of the analysis also served as an initial design phase for the interview questions, in line with the methodological approach previously outlined in Section 3. By analysing the visual representations, the authors' approach supported the creation of well-informed interview questions and helped ensure that the qualitative phase remained coherent with the quantitative data.

The thematic analysis of the interview data followed the structure presented by Braun and Clarke (2006) and was conducted using the digital tool Atlas.ti. The software was employed primarily to facilitate a shared coding approach among the authors, without utilising any of the AI functions embedded in the program. The initial analysis phase involved familiarising with the data, where interviews were transcribed, and the transcripts read multiple times to gain a deep understanding of the content. This was followed by the generation of initial codes, where data segments were identified as common patterns and relevant to the research question. These codes were created independently by two authors and then brought together in collaborative discussions. The purpose of these discussions was to critically evaluate the codes, address any assumptions underlying them, and begin to identify possible themes. As the final themes emerge from the interplay between the data, the analysis process, and subjectivity (Braun & Clarke, 2019), this collaborative approach helped maintain a balanced perspective, avoiding undue influence from individual subjectivity. The final step before producing the manuscript was properly naming the emerged themes to effectively convey the captured pattern observed within the data.

4. Quantitative results

The survey results are structured according to the AI literacy dimensions previously outlined in Section 3.1: Cognitive Dimension, Metacognitive Dimension, Affective Dimension, and Social Dimension. Additionally, a General Use category was included, which summarises the results from the frequency scale items that explore usage distribution across different activities and subjects, as well as other survey items that could not be related to students' AI literacy. A translated version of all the survey items discussed in the paper is presented in Fig. 3, which shows the distribution of the use of ChatGPT among STEM subjects; Fig. 4, which presents the distribution of ChatGPT use across different activities; and Table 2 reports the Likert-scale survey items discussed in the paper. The only exception is the item investigating the frequency of ChatGPT use, which was provided as an example in Section 3.3.

4.1. General Use

The survey results indicated that 88.4% of the students had used some form of LLM chatbot, such as ChatGPT, for their studies during the past month, with over 35% reporting a usage of more than five times within that period. This result, however, is general and not specifically related to STEM subjects. The survey item that explored students' frequency of ChatGPT use was phrased as described in Section 3.3. While a single use could potentially be interpreted as a prompt rather than a complete interaction, this aspect could affect the validity of the item. However, the second part of the question, which related each use to its support for student studies, provided sufficient context for participants to interpret the survey item properly.

Although the use of ChatGPT for studies was widespread among participants, only 20.4% agreed that ChatGPT can serve as an alternative to traditional learning materials, while over 63% disagreed. When examining which STEM subjects were most associated with using ChatGPT, biology emerged with the highest percentage of usage, followed by chemistry, as shown in Fig. 3, which shows the distribution of the use of ChatGPT among STEM subjects. More than 60% of participants reported using ChatGPT to explain concepts related to STEM subjects. This pattern may be connected to the fact that approximately 40% of participants used ChatGPT specifically for their biology studies, a subject characterised by numerous concepts. Apart from explaining concepts, the survey shows that most participants used ChatGPT for summarising texts (53%), reviewing material before tests (45%), and for school projects (40%), as shown in Fig. 4. The survey also included items that allowed students to specify additional uses of ChatGPT within STEM subjects through an optional input field. Although the number of students who provided this extra input was relatively small (n = 44), several recurring activities were mentioned, such as using ChatGPT for inspiration, programming, brainstorming, and even cheating.

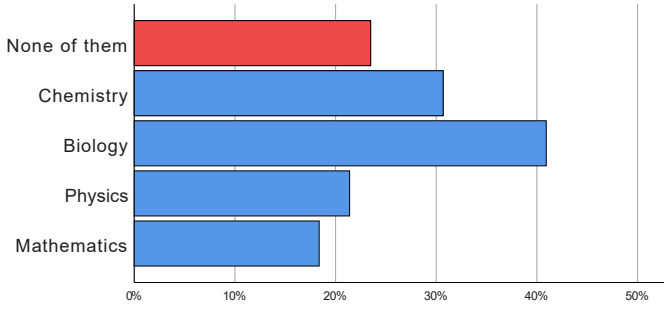


Fig. 3. Distribution of the use of ChatGPT among STEM subjects.

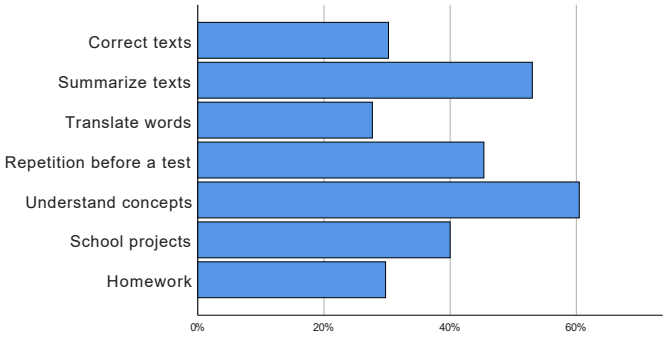


Fig. 4. Distribution of ChatGPT usage across different activities.

Table 2
Likert-scale survey items.

Survey item	Disagree	Neutral	Agree
General Use			
AI tools such as ChatGPT help me understand difficult concepts in the STEM subjects[a]	18.3%	9.7%	65.3%
AI tools such as ChatGPT can be a substitute to traditional learning materials such as books	63.8%	15.8%	20.4%
Cognitive dimension			
I have a good understanding of how AI tools such as ChatGPT generate texts	18.8%	17.4%	63.8%
I am familiar with the concept of machine learning, which is the technology behind AI tools such as ChatGPT	31.7%	19.1%	49.2%
Metacognitive dimension			
I can detect potential inaccuracies in responses generated by AI tools such as ChatGPT	31.7%	19.1%	49.2%
I double-check the information generated by AI tools like ChatGPT against other sources	20.8%	19.6%	59.6%
I trust the answers that AI tools like ChatGPT generate	33.2%	24.1%	42.7%
I adjust my questions to AI tools such as ChatGPT depending on the STEM subject I am working with[a]	11.7%	17.5%	61.0%
Affective dimension			
I am interested in learning more about how AI tools such as ChatGPT can be used in a way that helps me for my studies	15.3%	20.0%	64.7%
I feel that I have sufficient knowledge to use AI tools such as ChatGPT to support my studies in the STEM subjects	17.5%	20.0%	62.5%
Social dimension			
The use of AI tools like ChatGPT in schools will increase in the coming years	4.6%	11.0%	84.4%
I am concerned about privacy aspects when using AI tools such as ChatGPT for my studies	47.1%	30.2%	22.7%
I believe it is important to discuss how AI tools such as ChatGPT can be used in schools.	12.0%	19.0%	69.0%
I think the students at my school follow the rules regarding the use of AI tools like ChatGPT[a]	29.4%	29.2%	24.1%

Note. In the survey, the STEM subjects were specified. In the table, the "Disagree" category represents the combined percentage of responses for "Somewhat Disagree" and "Strongly Disagree," while the "Agree" category represents the combined percentage of responses for "Somewhat Agree" and "Strongly Agree."

[a]The percentage does not add to 100% since the students could also answer "I don't use any AI tool for my studies" or "I don't know if my school has any rules regarding the use of AI tools."

4.2. Cognitive dimension

The findings reported in Table 2, which presents a list of all the Likert-scale survey items, show that the participants were confident in their knowledge of specific technical concepts, such as machine learning and how ChatGPT generates texts. As much as 63.8% of the participants

felt confident in their knowledge about how ChatGPT generate responses, and interesting results since only 36.7% of the students agreed to the statement that the responses generated by ChatGPT can show some form of bias.

4.3. Metacognitive Dimension

Almost half of the participants (49.2%) reported confidence in identifying potential inaccuracies in the responses generated by ChatGPT, and 59.6% indicated that they actively double-checked the generated responses with different sources. This finding is particularly interesting in relation to the items exploring participants' trust in the content generated by ChatGPT, where only 33.2% reported not trusting compared to the 42.7% that reported trusting or partially trusting the accuracy of the responses. An investigation into a potential correlation between trust, the ability to identify inaccuracies, and the habit of double-checking was conducted, but the results were inconsistent. Additionally, most participants (61.0%) reported adapting their prompts according to the specific STEM subject, suggesting that some form of prompting strategy is being implemented in their interactions with ChatGPT.

4.4. Affective Dimension

A significant majority of respondents (62.5%) indicated confidence in their ability to use ChatGPT for their STEM studies. However, it is important to consider that the survey item was phrased as, "I feel that I have sufficient knowledge to use AI tools such as ChatGPT to support my studies in STEM subjects," which may have been interpreted as referring to their subject related content knowledge rather than their knowledge about AI. Additionally, 64.7% of participants expressed an interest in learning more about LLM chatbots to better support their education, while only 15.3% did not feel the need for further knowledge. This interest in learning more about AI aligns with the results of another survey item exploring students' beliefs about the future of AI usage, where 84.4% indicated that they believe the use of AI tools such as ChatGPT in schools will increase over time.

4.5. Social Dimension

The survey results showed that most participants (69%) consider it important to discuss regulations in school regarding how ChatGPT can be used. However, only 37% of students agreed that the use of ChatGPT was regulated in their schools. Most participants were either neutral (30.2%) or unconcerned about privacy aspects while using ChatGPT for their STEM studies (47.1%). It is important to note that the prompts used in this context usually do not contain personal information, so these results cannot be generalised to determine whether students are concerned about privacy when using these tools for other purposes. Additionally, 29.4% of respondents indicated that students in their school did not follow the existing regulations, in contrast to the 24.1% who believed that students did follow such regulations. It is worth mentioning that this survey item included the option to answer, "I am not aware of any regulations about the use of AI tools such as ChatGPT in my school," which explains the relatively low cumulative percentage.

5. Qualitative results

In this section, the themes resulting from the analysis are presented. Three themes were formulated: *Perceived role of ChatGPT in STEM learning, Understanding of how ChatGPT works* and *Attitudes towards ChatGPT*. The themes are divided into subthemes to further structure the results based on the qualitative data, as shown in Table 3 which shows the structure of themes and subthemes that emerged from the thematic analysis.

Table 3
Structure of themes and subthemes from the thematic analysis.

Theme	Subtheme
Perceived Role of ChatGPT in STEM Learning	• ChatGPT to understand and review STEM concepts • Calculations as a limitation of ChatGPT in STEM • Using ChatGPT effectively requires content knowledge
Understanding of how ChatGPT works	• Prompting Strategies • Misconceptions about ChatGPT
Attitudes towards ChatGPT	• Initial scepticism to adopt ChatGPT • Limited trust toward ChatGPT-generated content

5.1. Perceived role of ChatGPT in STEM learning

This theme emerged from students' descriptions of their experiences of ChatGPT's role in their STEM-related learning, such as its specific use for certain subjects and perceived challenges related to its adoption.

5.1.1. ChatGPT to understand and review STEM concepts

Participants experienced ChatGPT as a useful educational resource to understand various concepts in STEM subjects, particularly biology and chemistry. Compared to other digital resources like Google, one of the most reported benefits of using ChatGPT is its ability to respond to context-specific follow-up questions and clarify misunderstandings.

> Student B: I think it's very good at that, and I often use it for that, to understand what they [different words or concepts] mean, so I think it's often better at it than if you were to look it up elsewhere, perhaps, you might not get the correct definition that you would get from the National Encyclopaedia, but I think it explains things in an understandable way.

The excerpt highlights how students perceive the difference between exploring concepts with ChatGPT compared to other sources, recognising the potential for inaccuracies while appreciating the tool's ability to provide understandable and easily digestible content.

> Student G: AI explains things to me much better than, for example, the book. […] The AI provides the information faster. I don't have to go anywhere, but for example, if I had googled, I would have to visit a website and read through it to find the information.

Student G also finds ChatGPT more effective than other educational resources in explaining STEM concepts. In this case, Student G experiences the tool as useful primarily due to its efficiency rather than its accuracy or the quality of the generated content. Compared to other educational tools, the efficiency of ChatGPT was similarly noted by other participants, making it one of the key reasons for the perceived usefulness of this technology in explaining various STEM concepts.

Participants' experiences of ChatGPT's ability to explain various STEM concepts can also be observed through the usage descriptions provided. A recurring activity mentioned was using the technology to review concepts or specific terms before biology or chemistry tests. In this case, ChatGPT was either used to generate questions or to provide glossaries with requested terms. These glossaries were later used as study materials for review before the test.

> Student E: Before the first biology test, there were many terms when we were working on cell division and all that. So, there were many difficult terms that I asked [ChatGPT] about to understand what they meant, and then I got an explanation. That made it much easier for me to learn.

Student E describes how they used ChatGPT to reinforce their understanding of biology concepts, which helped them prepare for their

first major biology test. Other participants shared similar experiences in their STEM studies, particularly in biology and chemistry. While using these tools to support learning in biology was a common theme, not all students agreed on which subjects these tools were best suited for. Some participants found ChatGPT most beneficial in chemistry, while others believed it more helpful in physics or mathematics. Students' rationale for the perceived usefulness of the technology in specific subjects was more closely tied to their particular uses and prompting strategies rather than an understanding of ChatGPT's limitations. This aspect became evident when participants were asked to elaborate on their reasoning, as their responses were mostly focused on the quality of the generated answers.

5.1.2. Calculations as a limitation of ChatGPT in STEM

Despite differing views on ChatGPT's suitability across subjects, most students described the technology as less effective for tasks involving calculations. When asked about the limitations of ChatGPT for STEM studies, "mathematics" or "calculations" were the most common responses. Many students provided examples where ChatGPT's generated responses differed significantly from those in their study materials. For instance, Student J remarked, "I mean, I wouldn't use ChatGPT to figure out $1 + 1$, that's what a calculator is for, right?" This statement reflects Student J's limited trust in ChatGPT's' ability to perform even the simplest mathematical calculations. Because of this perceived limitation, some participants found ChatGPT less useful for mathematics than other subjects.

Although there were extensive concerns about ChatGPT's limitations in handling calculations, two of the interviewed students still found the technology to be helpful in supporting their learning of mathematics. For example, Student L shared:

> Student L: I didn't understand functions in mathematics at all for several months, but then eventually, I was like, "OK, I need to learn this sooner or later." So, one day, I sat down and studied math the whole day. I used ChatGPT, saying, "OK, explain functions," and when I didn't understand what it meant, I asked it to explain it in another way until I finally got it.

In this case, Student L did not use the tool to solve exercises requiring calculations but rather to grasp a concept they had struggled with. This example suggests that the standard view of ChatGPT as being less useful for mathematics is more related to its limitations in solving exercises rather than its overall utility in understanding mathematical concepts. A similar limitation was also noted in other STEM subjects, such as physics and chemistry, in the case that the topic required some form of calculation. Some participants described this limitation as diminishing the overall usefulness of ChatGPT for STEM subjects. These participants felt that the primary use of ChatGPT in education was limited to helping with homework or assignments, and the perceived inability of the technology to solve exercises made it ineffective for STEM subjects in general.

5.1.3. Using ChatGPT effectively requires content knowledge

Participants expressed that one of the key factors for effectively using ChatGPT in STEM studies is having some level of understanding of the subject matter to which the tool is being applied. They were aware of the potential for hallucinations generated by ChatGPT and recognised that content knowledge is essential for identifying these inaccuracies.

> Student H: If I don't know much about the subject, then I can get kind of lost. But if I have some idea of what it's about, I can see because it shows the steps it takes to solve tasks and such. Then I can just say, "What did it do there? That wasn't quite right," and I go through it. If it's made small mistakes, I can catch it.

In this example, Student H highlights the challenge of assessing the accuracy of a generated response without prior knowledge of the content. While the participant emphasised the importance of double-checking answers, this was typically in response to perceived inaccuracies, which they were able to identify due to their understanding of the topic being studied. This aligns with the earlier discussion on perceived effectiveness: if students lacked the ability to detect potential errors and always had to double-check the generated content, ChatGPT would not be seen as more effective than other educational resources.

Similarly, some participants attributed their ability to assess and evaluate ChatGPT-generated responses to a form of "intuition" or "gut feeling." Even in these cases, participants referred to having some level of content knowledge that helped them evaluate the quality of the generated content. They described reading a response and experiencing the feeling that something was not quite right, a reaction that still relies on an underlying understanding of the content knowledge.

5.2. Understanding of how ChatGPT works

This theme emerged from students' experiences with ChatGPT in relation to their prior knowledge of the technology. Participants' descriptions revealed that their understanding of this technology significantly influenced their use of ChatGPT. While not always explicitly stated, their underlying knowledge shaped how they experienced ChatGPT as a tool to support their STEM studies.

5.2.1. Prompting strategies

The interview data provided examples that explored the specific strategies adopted by the participants to generate responses within the context of STEM education. The complexity of these structured prompts varied among participants, but most demonstrated some form of strategy related to either the topic being explored or the activity for which the ChatGPT was being used. A recurring strategy was to explicitly specify the desired output, such as asking the tool to present the response in bullet points. In some cases, the desired output was only explicitly requested if the initial response was unsatisfactory.

Participants also shared specific words they used to shape the desired output. For example, Student F explained: "I usually use it like, if I want a specific text, I just write 'brief,' 'longer,' or 'medium.'" In other instances, the specification of the desired output was more complex and reflected a deeper understanding of how to use the technology, as exemplified by Student I: "I usually ask leading questions because then you get different answers, or if you ask, 'What is good about this?' you'll still get an answer about what's good." Student I's approach demonstrates a strategy to work around the potential bias that can characterise ChatGPT's responses. By framing questions to elicit both positive and negative aspects of a topic, the student is able to mitigate potential bias and obtain more balanced responses.

Another strategy employed by participants involved providing ChatGPT with sufficient context to generate more accurate and relevant responses. The level of contextualisation varied, ranging from specifying the STEM subject or the specific topic within the subject to even providing examples of previously solved tasks. One participant described how they contextualise their prompt by including specific facts they want ChatGPT to incorporate into the response:

> Student J: If there's some fact that I want it to bring up, I include that in my prompt. Like, "blah blah blah, Sweden has a smaller population than Australia," and then that it should integrate it into the response.

In this last example, Student J showed a strategy to avoid potential hallucinations by providing the tool with the required data and only using ChatGPT's linguistic capabilities.

5.2.2. Misconceptions about ChatGPT

Some of the interview questions were also formulated to explore participants' general understanding of AI concepts such as machine learning and their understanding of LLM chatbots. Participants showed little to no understanding of machine learning except for one of the

twelve participants. Additionally, when asked the question "How do you think that AI tools like ChatGPT generate responses?" almost all the participants showed some form of misconception and mainly described LLM chatbots as search engines.

> Student B: When I write a question about biology on a certain topic, it might look for matches on the internet, because I assume it bases its answers on what's already available out there. It's good at pulling from different sources and summarising the information.

Student B, like most participants, believed that ChatGPT actively searches for information on the internet to generate responses and did not mention any form of pretraining in the process. Other forms of misconceptions were, for instance, the idea that the models are constantly being trained on user prompts in real time or the assumption that ChatGPT can only generate biased content about topics related to politics. This aspect is evident in the following excerpt from Student C's interview: "[…] I've seen on social media where people have asked sensitive questions about Donald Trump or Joe Biden, and the answers have been biased, for sure. But yeah." When later asked if this perception influenced how they use the tool for their STEM studies, Student C responded, "No, not at all. I don't think it does. I haven't used it for anything political or anything like that." From this excerpt, it is clear that Student C is aware of the potential for bias in generated responses but believes this issue to be only related to topics like politics and not applicable to STEM subjects. Similarly, most participants acknowledged the potential for bias in ChatGPT's responses but, in most cases, did not consider it relevant to the use of the technology for STEM education.

5.3. Attitudes towards ChatGPT

Finally, the last theme emerged from the codes reflecting participants' descriptions of their attitudes towards adopting ChatGPT for their STEM studies. Although participants' experiences varied, the interview data revealed recurring patterns in their processes of initial adoption and thoughts about ChatGPT-generated content.

5.3.1. Initial scepticism to use ChatGPT

Five out of twelve participants were initially sceptical of ChatGPT in general and only began using the technology for their studies after some time. This initial scepticism stemmed from concerns about cheating, violating ethical standards, or not completing the work independently.

> Student K: I was a bit against it at first, but then I realised that AI keeps developing, and it's here to stay. So, I think you really must use it and, well, take advantage of it instead of just focusing on the negatives. Of course, there are a lot of negative aspects with AI, but the only way to try to develop and move forward is to use and improve that ability yourself. Otherwise, you'll probably fall behind others who are using AI.

Although initially sceptical, Student K's reasoning for adopting ChatGPT shifted from practical use to the importance of AI literacy as a set of necessary skills to remain competitive. Other participants reported this aspect, emphasising that developing AI literacy would make them more likely to secure future job opportunities. Many described using ChatGPT for their studies, not just in STEM subjects, as an effective strategy for building this skill. Most participants reported that their ability to generate helpful and relevant responses improved over time, indicating that practice plays a significant role in developing proficiency with ChatGPT.

5.3.2. Limited trust toward ChatGPT-generated content

A significant majority of the participants (n = 11) expressed limited trust in the content generated by ChatGPT. While the level of trust varied among them, the general sentiment was that the generated content could not be fully trusted, and important information needed to be verified using other sources. The more trustworthy resources mentioned included teachers, reliable websites, and books. This lack of trust extended beyond calculations to content in general, reflecting students' experiences with potential hallucinations in generated responses. Another factor contributing to this limited trust was the absence of references in ChatGPT-generated content, which some students found "scary." Additionally, concerns were raised about these technologies being privately owned by companies that could potentially impose their own agendas.

> Student C: For example, when I need to write a short paragraph, I let it write the paragraph, then I read through it and think, "Yeah, this makes sense." But if I'm looking for facts, I'd rather use a search engine like Google or something similar.

As other participants also mentioned, Student C does not trust ChatGPT to gather facts about scientific content. In cases where the technology was used for activities requiring gathering specific facts, the information was either provided by the student or double-checked afterwards.

> Student J: I could have trusted it a lot more if it had the ability to acknowledge when it's uncertain or if there are a lot of conflicting answers. If that feature existed, I could trust it much more. But because that information isn't available to us as users, it's just … well, it's limited.

Student J takes the trust argument a step further, emphasising that ChatGPT-generated content cannot be fully trusted because the technology does not typically explain whether the sources behind the generated content contain contradictory information, making it difficult to summarise accurately in a response.

6. Discussion

The growing need for STEM workers to support societal development, combined with the increasing adoption of LLM chatbot technologies by students, formed the basis for this exploratory study. The main goal of the mixed-methods approach was to gather data to better understand students' perceptions and use of ChatGPT in STEM subjects, as well as their knowledge of AI. The quantitative data focused on students' AI literacy and specific usage patterns, while the qualitative data aimed to uncover the reasoning behind particular use strategies. This approach sought to explore how upper secondary students' understanding and perceptions shaped their experiences with ChatGPT in the STEM context.

As suggested by the results of the quantitative part of the study, a majority of participants found ChatGPT particularly useful for developing their understanding of concepts within STEM subjects. The interview data shed further light on this aspect, showing that this result was mainly due to the ChatGPT's ability to deliver quick answers and engage in conversations characterised by specific follow-up questions. This finding aligns with the work of Liang et al. (2023) who argue that ChatGPT can provide a safe environment for students to ask questions without worrying about revealing gaps in their understanding to teachers or classmates. Student L's description of their process for understanding mathematical functions, as reported in Section 5.1.2, supports this argument. Having an extended conversation with ChatGPT helped the student to finally grasp a topic they had been struggling with for a long time. This example illustrates how ChatGPT can be used to answer specific, contextualised questions in STEM within an environment where time for individual queries is virtually unlimited, emotional pressures are minimised, and the teacher's attention is not divided among other students. The perceived societal importance of STEM education underscores the need to explore this topic further, particularly to understand how such a safe environment can support student learning in STEM subjects.

The survey results showed that respondents used ChatGPT more frequently in biology and chemistry than mathematics and physics,

which can be explained by the fact that participants struggled to use these tools for tasks requiring numerical calculations. Although physics is also characterised by numerous concepts, students reported during the interviews that difficulties using the technology for the mathematical aspects of the subject, which shaped their overall experience. The poor performance of LLM chatbots on numerical tasks has been well-documented, even before the release of ChatGPT (Cobbe et al., 2021). Despite the fact that the performance has improved for these particular tasks over the years (Polverini & Gregorcic, 2024), it is not surprising that students continue to find the technology challenging to use in calculation-heavy subjects. However, an alternative argument could be made: subjects involving numerical calculations often come with clear answer sheets, allowing students to quickly compare ChatGPT-generated responses with the correct solution. In such cases, it is easier for users to spot inaccuracies or eventual hallucinations. In contrast, subjects like biology, where answer sheets are less commonly provided in the same way, make it more challenging to detect eventual errors.

The presented data also provides evidence that technological knowledge about AI may not be a prerequisite for developing effective prompting strategies. Although only one of the interviewed students demonstrated specific knowledge about topics such as machine learning, most students provided several examples of effective prompting strategies. The strategies described by these students, such as contextualising the prompt and specifying the domain, align with those outlined by Polverini and Gregorcic (2024). However, the results also suggest that while technical knowledge of AI technologies may not be essential for creating effective prompts, awareness of potential bias and hallucinations can still help students develop effective prompting strategies and better assess which content is best suited for ChatGPT. This result is in line with the findings presented by Knoth et al. (2024) who argued that although specific knowledge about AI may be beneficial to the development of prompting strategies, AI literacy might not be a general requirement, especially the competencies heavily related to the technical part of the technology. This aspect of the discussion can be considered in relation to the generative AI literacy model presented by Annapureddy et al. (2024) and used to structure the quantitative results of these studies. According to the students' experiences analysed in this study, specific competencies such as *Knowledge of generative AI models* were not indicative of the participant's ability to use the technology in a way that was perceived as effective to support their STEM studies, nor did it limit students' ability to develop effective prompting strategies over time. Furthermore, the students who reported detailed prompting strategies also showed a more positive attitude toward ChatGPT and argued for its usefulness, which aligns with the results presented by Tassoti (2024) and Theophilou et al. (2023). This aspect of the results supports the argument that developing effective prompting strategies is a key factor that shapes the overall experience of ChatGPT for educational practices.

Participants regarded content knowledge as one of the key factors in assessing the quality and relevance of generated content. This aspect aligns with the key components for prompting presented by Cain (2024), where Content knowledge is described as one of the key factors in creating effective prompts. Although more than 40% of participants reported trusting the generated responses, they still viewed subject-related content knowledge as crucial for critically evaluating the accuracy and quality of the information. This finding connects to broader discussions about the risks of overreliance on AI tools and the potential impact on critical thinking skills (Chavanayarn, 2023; Dwivedi et al., 2023; Koskinen, 2024), where participants experienced that content knowledge remains the most crucial factor, rather than other competencies associated with AI literacy models.

6.1. Implications for STEM learning

The emphasis on content knowledge raised by participants highlights a critical aspect of ChatGPT's role in educational practices. While participants identified several potential uses for this technology in STEM studies, it is essential to ensure that students possess sufficient foundational knowledge about the concepts to which ChatGPT is applied. This is particularly important to avoid misconceptions that may arise from generated hallucinations, which could negatively impact students' learning processes. The risk of reinforcing or introducing additional misconceptions could affect students' knowledge base and hinder their ability to build on this knowledge effectively. Such outcomes could have significant implications for students' continuous learning and long-term academic progress (Liu & Fang, 2016). However, as suggested by the findings, if the technology is used with sufficient understanding of the content knowledge, it could support the learning of STEM subjects by scaffolding students, for example, by providing additional exercises or answering questions (Sharma & Hannafin, 2007).

The increasing use of this technology by students in STEM education shown in the result, raises intriguing questions when analysed through the lens of a sociocultural perspective on learning. In this context, learning is understood as a phenomenon situated within a context where mediating cultural tools, such as language, play a pivotal role for students' learning process (Säljö, 2021). This phenomenon can be generalised across all subjects, including, of course, STEM disciplines. What are the implications if students are increasingly exposed to text not produced by humans but by LLM chatbots such as ChatGPT? How might this phenomenon shape students' understanding of complex concepts, and how could it affect their comprehension in the long term? While the findings presented in this paper provide insights into students' perceptions of the role this technology plays in their learning of STEM subjects, they do not address the potential long-term implications that this interaction could have on students' learning process.

6.2. Limitations and future studies

Some of the limitations of this study were the focus on only a few subjects under the STEM umbrella and conducting a limited number of interviews. Additionally, the exploratory approach touched on various aspects of students' experience of ChatGPT without delving deeply into specific implications. Another limitation of this study is that it explored students' experiences with ChatGPT while possibly prompting in Swedish, a language underrepresented in the model's training data. This limitation could impact the generalisability of the results due to the technology's reduced proficiency in this particular language.

Future research should, for instance, examine specific phenomena highlighted in this paper, such as the role of AI literacy in shaping students' prompting strategies and the relationship between trust in generated responses and content knowledge. Additionally, as shown in the results, the substantial use of ChatGPT to understand concepts underscores the importance of conducting further studies to investigate the implications of students adopting ChatGPT for understanding STEM concepts.

7. Conclusions

This study aimed to explore upper secondary students' experiences with ChatGPT in STEM education, focusing on their perceptions, use, and specific knowledge about AI. The results show extensive adoption of LLM chatbots among participants, with varying outcomes across different STEM subjects. Biology and chemistry had the highest adoption rates, mainly because of students' experience of ChatGPT as a valuable supplement for understanding concepts. At the same time, the technology was experienced as less effective for numerical calculations. Although participants lacked in-depth technical knowledge about AI, they demonstrated effective prompting strategies that developed over time and influenced their overall experience with ChatGPT for STEM studies.

While ChatGPT has been studied outside of the STEM education

context, there is a pressing need for more empirical studies that investigate the role of LLM chatbots in STEM subjects, particularly at the upper secondary education level. This could help develop educational practices that better support students in navigating these complex subjects. Given the increasing need for STEM workers and the consequential need to encourage more students to pursue STEM studies, it is essential to explore tools that can support them effectively, especially as these technologies continue to be adopted regardless of formal guidance.

CRediT authorship contribution statement

Federico Valeri: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pernilla Nilsson:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Anne-Marie Cederqvist:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Data availability

The data can be obtained by sending request e-mails to the corresponding author.

Ethics statement

The study was conducted in full compliance with relevant laws, guidelines, and regulations of Halmstad University, and approval was received from the university's internal board. Informed consent was obtained from all participants, and their privacy rights were rigorously upheld.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT-4o to increase the readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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